

Mukesh Kumar Pandey

LinkedIn: [linkedin.com/in/mukesh4783/](https://www.linkedin.com/in/mukesh4783/)
 GitHub: github.com/mukesh4783

Email: mukeshkumarpandey82@gmail.com
 Mobile: +91-84240 76924

Skills

Languages: Python, C/C++, Java
Tools & Frameworks: Github, PowerBI, FastAPI, Flask, Streamlit, Scikit-learn, Matplotlib, Seaborn, SciPy, Manim
Databases: MySQL, PostgreSQL
Soft Skills: Problem-Solving, Time Management, Multitasking, Self-Learning.

Experience

Inventory and Biling Managemnet System
Jun 2025 – Jul 2025

Developed a Python & MySQL-based Inventory and Billing Management System with separate customer and manager modules, featuring real-time stock updates, product management, pricing, and automated bill generation.

Projects

WebRAG — Grounded Q&A Bot | Github
Jul 2026

- Built a Retrieval-Augmented Generation app that answers questions strictly from live, user-supplied web pages instead of the model's training memory.
- Engineered a LangChain pipeline (WebBaseLoader, RecursiveCharacterTextSplitter with 1000-char chunks/200-char overlap) feeding OpenAI's text-embedding-3-small and GPT-4o-mini into a local Chroma vector store, with SHA-256 hash comparison to auto-detect page changes and a Streamlit UI showing a diff-based freshness report.
- Cut hallucinated responses to near 0% on out-of-scope queries, improved retrieval relevance by 30%, and reduced per-query latency by nearly 40% versus cloud-hosted vector DBs.
- Tech used: LangChain, Chroma Vector Database, OpenAI Embeddings, GPT-4o-mini, Streamlit, Python

GramSetu — Digital Village Management Portal | GitHub
Jul 2026

- Architected a full-stack digital governance portal bridging rural administration and citizen services online.
- Developed citizen-facing modules for service requests, welfare scheme applications, and certificate issuance with real-time tracking, an admin analytics dashboard (Chart.js), a multilingual Google Gemini chatbot, and deployment on Vercel with MongoDB Atlas and JWT-based role auth.
- Digitized 10+ citizen workflows, improved service-request processing visibility by roughly 45%, and scaled to 500+ concurrent sessions at 99% uptime.
- Tech used: React, Node.js, Express.js, MongoDB Atlas, Chart.js, Google Gemini API, Vercel

Manimax | Github
Apr 2026

- Established an AI-powered pipeline that converts natural language prompts into fully automated educational video content.
- Programmed automated Manim script generation using local LLMs to eliminate dependency on external AI APIs, rendering precise mathematical animations, equations, and synchronized narration text, and containerized the rendering engine and local LLM with Docker for a portable, zero-dependency deployment.
- Cut manual animation scripting time by nearly 70%, reduced per-video inference cost by 60%, and cut environment setup time by 50%.
- Tech used: Python, Manim, Docker, Local LLMs, AI Pipeline

Training

AI Engineer Launchpad Mastering LLMs and Agentic AI | Certificate
May 2026 – Jun 2026

- Developed a strong foundation in LLMs and Transformer architectures, covering attention mechanisms, tokenization, embeddings, and pre-training objectives.
- Applied advanced prompt engineering and reasoning techniques, including zero-shot, few-shot, chain-of-thought, structured prompting, and function calling.
- Created RAG applications using LangChain, ChromaDB, and FAISS, implementing document chunking, semantic search, indexing, retrieval, and query transformation.
- Gained hands-on experience with LoRA and QLoRA fine-tuning, quantization, instruction tuning, and evaluation of fine-tuned language models.

Certificates

Oracle Cloud Infrastructure AI Foundations Associate | Oracle
 Jul 2026

Database Management System | Infosys
 Aug 2026

Programming Using C++ | Infosys
 Aug 2025

Achievements

Placed Top 15 in Resurgence Hackathon at Lovely Professional University
 Apr 2026

Achieved B certificate with A grade in National Cadet Corps
 May 2024

Education

Lovely Professional University
Phagwara, Punjab
 Bachelor of Technology - Computer Science and Engineering; CGPA: 8.89
 Aug 2024 – Present

Army Public School
Beas, Punjab
 Intermediate; Percentage: 86%
 Apr 2023 – May 2024

Army Public School
Beas, Punjab
 Matriculate; Percentage: 92.6%
 Apr 2021 – May 2022